



Motivation

- Distributed optimization
- Worst-case, non-smooth
- Coupling of constraints

Constrained Minimax Optimization

$$\mathbf{f} : \Theta \rightarrow \mathbb{R}^{|\mathcal{I}|} \text{ and } \mathbf{g} : \Theta \rightarrow \mathbb{R}^{|\mathcal{I}|}$$

$$\begin{aligned} \min_{\mathbf{w} \in \Theta} F(\mathbf{w}) &= \min_{\mathbf{w} \in \Theta} \max_{i \in \mathcal{I}} [\mathbf{f}(\mathbf{w})]_i = \min_{\mathbf{w} \in \Theta} \max_{i \in \mathcal{I}} f_i(\mathbf{w}) \\ \text{s.t. } G(\mathbf{w}) &:= \max_{i \in \mathcal{I}} [\mathbf{g}(\mathbf{w})]_i = \max_{i \in \mathcal{I}} g_i(\mathbf{w}) \leq 0 \end{aligned}$$

Softmax-Weighted Switching Gradient

Switching Condition

Check if $G_k(\mathbf{w}_k) = \langle \text{softmax}(\alpha \mathbf{g}(\mathbf{w}_k)), \mathbf{g}(\mathbf{w}_k) \rangle \leq \frac{\epsilon}{2}$

Update Rules

- ▶ If $G_k(\mathbf{w}_k) \leq \frac{\epsilon}{2}$ (constraints satisfied):
 - ▶ Recompute $\mathbf{p}_k \leftarrow \text{softmax}(\alpha \mathbf{f}(\mathbf{w}_k))$
 - ▶ Update direction: $\mathbf{u}_k \leftarrow (\nabla \mathbf{f}(\mathbf{w}_k))^\top \mathbf{p}_k$
- ▶ Else (constraints violated):
 - ▶ Recompute $\mathbf{p}_k \leftarrow \text{softmax}(\alpha \mathbf{g}(\mathbf{w}_k))$
 - ▶ Update direction: $\mathbf{u}_k \leftarrow (\nabla \mathbf{g}(\mathbf{w}_k))^\top \mathbf{p}_k$
- ▶ Parameter update: $\mathbf{w}_{k+1} \leftarrow \mathbf{w}_k - \eta \mathbf{u}_k$

Return averaged solution: $\bar{\mathbf{w}}_K \leftarrow \frac{1}{|\mathcal{S}|} \sum_{k \in \mathcal{S}} \mathbf{w}_k$ where $\mathcal{S} \subset \{0, 1, \dots, K-1\}$ is the set of iterations where constraints are satisfied.

Optimality gap / feasibility tolerance

$$F(\bar{\mathbf{w}}_K) - F(\mathbf{w}^*) \leq \epsilon, \quad G(\bar{\mathbf{w}}_K) \leq \epsilon$$

Complexities

Method	Iteration Complexity (K)	Sample Size (B_ζ)	Total Oracle Complexity
[Lan and Zhou, 2020b]	$\mathcal{O}(\epsilon^{-2} \log^2 \frac{1}{\delta})$	$\mathcal{O}(\epsilon^{-2} \log^2 \frac{1}{\delta})$	$\mathcal{O}(\epsilon^{-4} \log^4 \frac{1}{\delta})$
Ours [Theorem 4.6]	$\mathcal{O}(\epsilon^{-2} \log \frac{1}{\delta})$	$\mathcal{O}(\epsilon^{-2} \log \frac{1}{\epsilon \delta})$	$\mathcal{O}(\epsilon^{-4} \log \frac{1}{\delta} \log \frac{1}{\epsilon \delta})$

High Prob. guarantee for Full Participation case (number of participating clients $m = n$, where n is the cardinality of $\mathcal{I}$)
To evaluate the function values of f_i and g_i , where the expectation $f_i(w) = \mathbb{E}[f_i(w, \zeta)]$ is empirically approximated by a sample average, a batch of B_ζ samples is required.

Error Decomposition

The optimality gap / feasibility tolerance ϵ is composed of three distinct error terms:

$$\epsilon \asymp \underbrace{\frac{DL}{\sqrt{K}}(1 + \bar{\sigma}_g^2)}_{\text{optimization error } \epsilon'} + \underbrace{\frac{\sigma_\zeta}{\sqrt{B_\zeta}}}_{\text{estimation error}} + \underbrace{\frac{\sigma}{|\ln(1-r)|n}}_{\text{sampling error}}$$

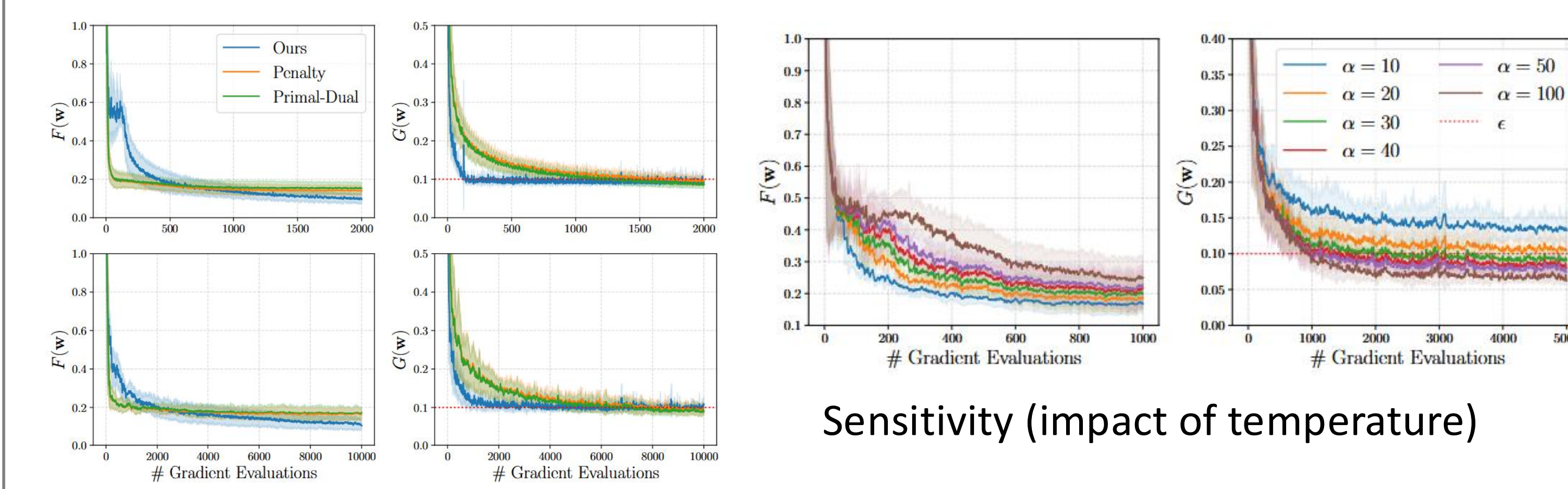
where $r := m/n$ is the participation ratio, and the remaining parameters are defined as:

- $\bar{\sigma}_g^2 := \frac{\sigma_g^2/B_g}{L^2 E}$ is **effective variance** in stochastic gradient approximation during optimization process, scaled by batch size B_g , number of local updates E , and L -Lipschitz constant.
- σ_ζ is associated with the **estimation** of expectation function values using a finite batch of B_ζ samples.
- σ quantifies the client **sampling noise** arising from the heterogeneity of function values across the population of n clients.

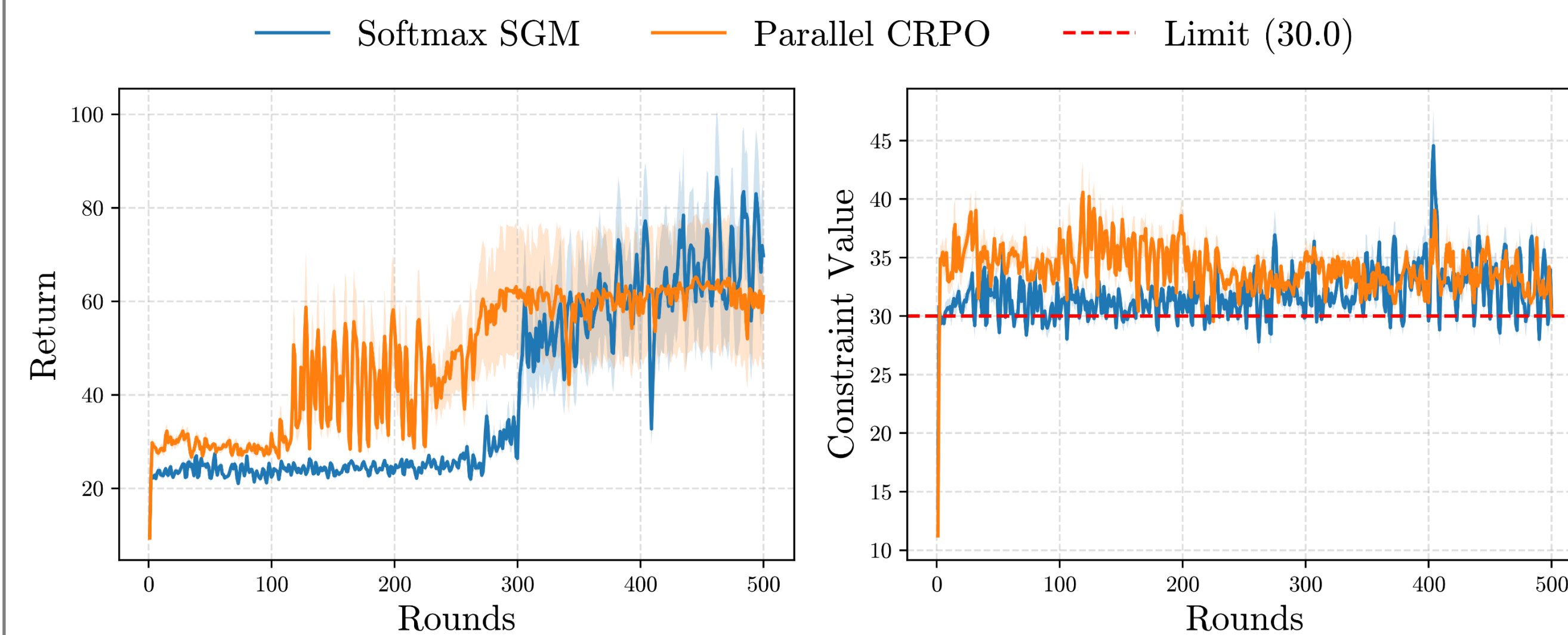
This bound is achieved with a smaller softmax hyper-parameter $\alpha \gtrsim \frac{\ln m}{\epsilon'}$, compared to $\frac{\ln n}{\epsilon'}$ required in the full participation case ($m = n$).

Experiments

NP classification



Federated Safe RL (Constrained MDP)



Stochastic Superiority via First-Order Stochastic Dominance (FSD)

Definition 4.8 (Stochastic Superiority via First-Order Stochastic Dominance (FSD), see [Shaked and Shanthikumar \[2007\]](#)). Let X and Y be two random variables. We say X is *stochastically superior* to Y in the sense of First-Order Stochastic Dominance (FSD), denoted by $X \succeq_{st} Y$ or $Y \preceq_{st} X$, if:

$$\mathbb{P}(X \geq t) \geq \mathbb{P}(Y \geq t) \quad \text{for all } t \in \mathbb{R}.$$

Assumption 4.9 (Uniformly Bounded Relative Gap). Let i be an index chosen uniformly at random from $\mathcal{I}$, and let the relative differences $D_{\mathbf{f}(\mathbf{w})} := F(\mathbf{w}) - f_i(\mathbf{w})$, $D_{\mathbf{g}(\mathbf{w})} := G(\mathbf{w}) - g_i(\mathbf{w})$. We assume that

$$D_{\mathbf{f}(\mathbf{w})} \preceq_{st} U, \quad D_{\mathbf{g}(\mathbf{w})} \preceq_{st} U$$

for $U \sim \text{Unif}[0, \sigma]$ with some $\sigma > 0$ and any $\mathbf{w} \in \Theta$.

Tighter Lower Bound for Temperature

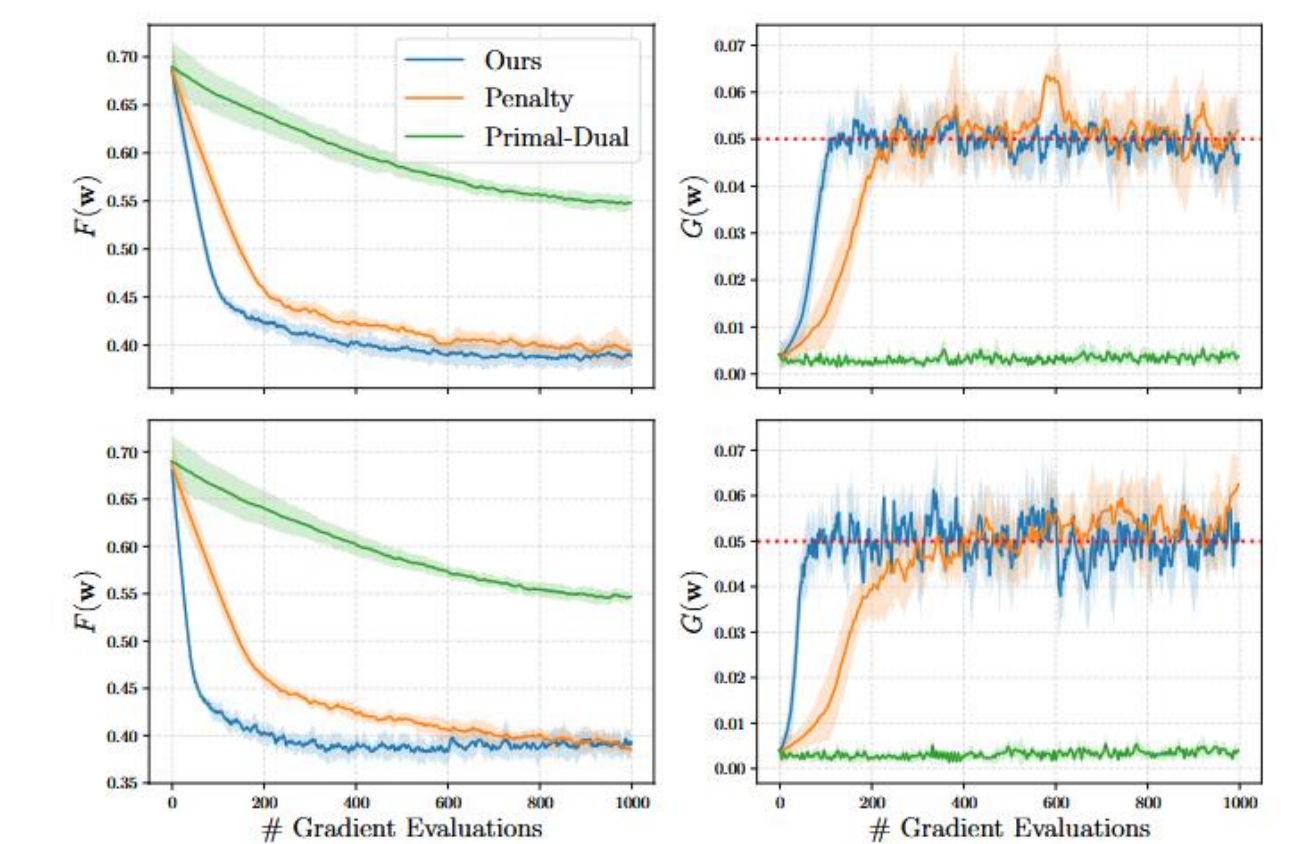
We achieve a tighter lower bound $\alpha \gtrsim \frac{\ln n}{\epsilon'}$

which improves upon [Wang et al., 2023]

$$\alpha \gtrsim \frac{1}{\epsilon'} (\ln n + \ln \frac{1}{\epsilon'} + \ln B)$$

by removing requirements [Wang et al., 2023] that functions be bounded between 0 and B.

Fairness Classification



Full participation (top: m/n=1)

Partial participation (bottom: m/n=0.5)

Setting	Return
Softmax SGM (ours)	69.73±67.42
Parallel CRPO	60.99±69.52
Setting	Violations
Softmax SGM (ours)	306.60±24.55
Parallel CRPO	379.20±24.59